

Memo 13: MP-only Numerical Limiter Update for `PLAST_FLOC=F` Baseline Stability in `mod_plast.F`

FVCOM-MP WaterPACT Investigation

May 25, 2026

Purpose

This memo records the model-development change made after the MP-only baseline runs with `PLAST_FLOC=F` showed numerical divergence. The main purpose of the change is to make the non-floc plastic pathway mirror the same water-column upper-bound value limiter already used by the split `PLAST_FLOC=T` pathway.

The project-level reference consulted for this memo was:

```
C:\Users\huan111\OneDrive-PNNL\Desktop\WaterPACT_Local\4_Floc_MP_
interaction\Memory\v003_mp_mass_divergence_diagnosis.tex
```

Background

The divergent cases were MP-only baseline tests using `PLAST_FLOC=F`. The reference memo showed that the river MP inputs were small and finite, while the model plastic mass grew to unrealistically large values during the run. The growth was therefore not consistent with a forcing-magnitude problem. Instead, the diagnostic checkpoints suggested that the non-floc transport pathway could create very large positive or negative combined plastic concentrations before the later lower-bound clipping stage.

The important numerical asymmetry was:

- With `PLAST_FLOC=T`, the split plastic fields `cnew_a` and `cnew_d` were already capped at 100 before the `CP7_after_upper_clamp` mass checkpoint.
- With `PLAST_FLOC=F`, the combined non-floc field `cnew` passed through advection and vertical diffusion without the corresponding upper-bound cap.

That difference made the nominal MP-only baseline a different numerical pathway, not simply a floc-disabled version of the coupled pathway. In particular, once very large negative values were generated, the later non-negative clamp could zero them while leaving compensating positive excursions, producing an artificial mass increase.

Code Change Recorded

The current development version of `mod_plast.F` adds an upper-bound limiter to the combined non-floc concentration:

where `(plast(iplast)%cnew > 100.0_sp) plast(iplast)%cnew = 100.0_sp`

This limiter is applied after the non-floc advection and vertical diffusion updates and before `CP7_after_upper_clamp`. The same treatment was added in both relevant non-floc branches:

- the sediment-enabled branch used by the main FVCOM-MP sediment build;
- the non-sediment fallback branch guarded by the preprocessor alternative.

The mass-check labels were also clarified in the non-floc branch. For `PLAST_FLOC=F`, checkpoints `CP3p5` through `CP6` represent the same post-transport, pre-cap combined concentration state. `CP7_after_upper_clamp` now records the state after the new upper-bound limiter.

Before and After

Aspect		Before	After
<code>PLAST_FLOC=T</code> path	split	<code>cnew_a</code> and <code>cnew_d</code> capped at 100 before <code>CP7</code> .	Unchanged.
<code>PLAST_FLOC=F</code> combined path	com-	<code>cnew</code> was advected and vertically diffused, but no equivalent upper cap was applied before <code>CP7</code> .	<code>cnew</code> is capped at 100 before <code>CP7_after_upper_clamp</code> .
Lower-bound treatment		The later non-negative clamp was already applied after open-boundary and point-source handling.	Unchanged. The new change only adds the missing upper-bound limiter.
Physical parameterization		No change to aggregation, disaggregation, settling, density, or floc coupling physics.	Still no change to those physics. This is a numerical value limiter.

Interpretation

This update makes the MP-only baseline numerically more comparable to the floc-enabled configuration. It does not turn on floc coupling, aggregation, disaggregation, equivalent grain size interaction, or sediment-floc physics when `PLAST_FLOC=F`. It only prevents the combined non-floc plastic field from exceeding the same upper concentration bound that already protected the split `PLAST_FLOC=T` fields.

The change should reduce the risk that an MP-only baseline diverges simply because it lacks a limiter present in the coupled run. Therefore, if the baseline becomes stable after this update, the result should be interpreted as a correction to the numerical pathway rather than a change in the physical meaning of the baseline case.

Recommended Verification Runs

For the next short P0/P1 verification, use the same forcing and model setup that previously diverged, but retain the plastic mass-check output. The key diagnostic comparison is:

- `CP3_after_update_bottom`: mass before water-column advection and diffusion;

- `CP3p5_after_adv_a` through `CP6_after_kinetics`: the post-transport, pre-cap combined non-floc state;
- `CP7_after_upper_clamp`: the first checkpoint after the new cap;
- `CP10_after_neg_clamp`: the state after the existing non-negative lower-bound clamp.

The expected healthy behavior is that `CP7` no longer permits extreme positive concentrations, and `CP10` no longer shows explosive mass growth caused by clipping very large negative values. If divergence persists, the next isolation tests should be a transport-only run with point sources disabled, followed by a focused search for the first cell and time step where `cnew` becomes non-finite or very large in magnitude.

Summary

The development repository now includes a missing upper-bound limiter for the combined MP concentration used when `PLAST_FLOC=F`. This brings the MP-only pathway into closer numerical alignment with the existing `PLAST_FLOC=T` split-field pathway and should make the baseline tests a fairer comparison for long-run MP-floc experiments.